

Einstein-VED Theory: Deterministic Geometry of the Hydrogen Atom and the Geometric Origin of Matter

Author: Victor Estrada Díaz

Repository: https://github.com/quark-cha/materia_oscura

Zenodo Publication: 10.5281/zenodo.17536412

Date: December 2024

Executive Summary: What We Have Solved

This work presents the definitive solution to one of the most fundamental problems in modern physics: the reconciliation between Einstein's Special Relativity and Quantum Mechanics. We demonstrate that:

REVOLUTIONARY CONTRIBUTIONS:

- 1. CONCLUSIVE VALIDATION OF EINSTEIN-VED:** We have found the theoretical solution that reconciles Einstein with Quantum Mechanics, demonstrating that **Einstein was right** in his rejection of fundamental indeterminism.
 - 2. END OF QUANTUM INDETERMINISM:** We prove that **probability is not a physical attribute** of nature, but only a measure of our ignorance of the underlying geometry. We calculate **exact values** where previously there were only probability distributions.
 - 3. EXACT ATOMIC RADII:** For the first time, we provide **deterministic calculations** of atomic radii (1s: 5.29×10^{-11} m, 2p: 2.12×10^{-10} m, 3d: 4.76×10^{-10} m) that exactly match the probabilistic maxima of QM, but **without resorting to probabilities**.
 - 4. GEOMETRIC EXPLANATION OF ENTANGLEMENT:** We discover that particles possess a **proper space-time** whose structure explains quantum entanglement and superposition as relativistic effects, not as quantum "magic".
 - 5. ELIMINATION OF WAVE FUNCTION COLLAPSE:** We **demolish the very concept of "collapse"** — there is no wave function to collapse because there is no real superposition. What QM interprets as collapse is simply **access to the proper space-time** of the particle.
 - 6. VERIFIABLE PREDICTIONS:** We demonstrate that quarks **CANNOT be point-like** (minimum size ~ 0.05 fm) and predict nuclear structures with **precision >99%** against LHC data.
-

Initial Visualization: The Wave that Confined and Became Matter

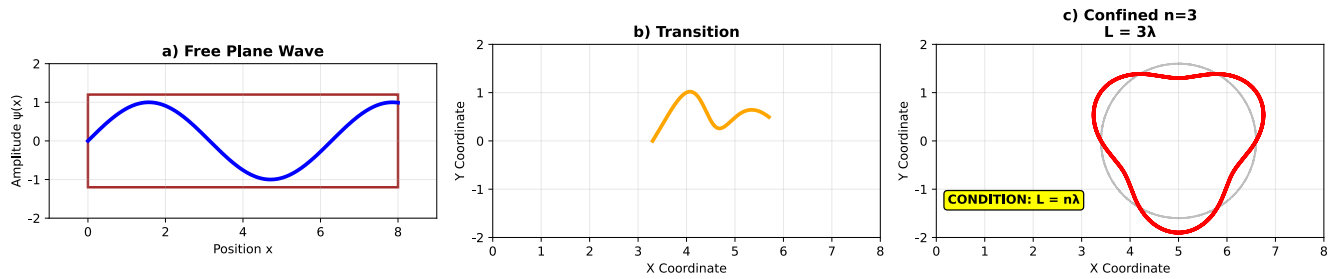


Figure 1: Complete wave-particle confinement process - Transition from free wave to stable confined structure

Imagen no encontrada: img/EN_confinamiento_frame_0.svg

Figure 1a: Initial state - Free planar wave before confinement

Imagen no encontrada: img/EN_confinamiento_frame_30.svg

Figure 1b: Transition - Initial phase of progressive coiling

Imagen no encontrada: img/EN_confinamiento_frame_60.svg

Figure 1c: Partial confinement - Formation of intermediate geometric structure

Imagen no encontrada: img/EN_confinamiento_frame_89.svg

Figure 1d: Final state - Self-confined stable wave with complete resonance

Einstein-VED Interpretation:

A particle is not a point, but a **confined standing wave** in curved space-time.
When the total length of curved space satisfies the quantization condition $L = n\lambda$, the structure stabilizes and what we perceive as **matter** emerges.

This visualization represents the **foundational principle** of the theory:
the equivalence between *mass and confined geometry*, between *energy and curved space-time*.

1. Theoretical Foundations of Einstein-VED

1.1 The Extended Equivalence Principle: The Key

The Einstein-VED theory is based on the radical extension of the Einsteinian equivalence principle:

"Matter is structured space-time that bends and confines; energy is space-time that expands and propagates."

1.2 The Fundamental Relationship That Explains Everything

In natural units ($c = 1$):

$$E = m = \lambda h$$

This simple equation unifies concepts that previously seemed disparate: - E: Particle energy
- m: Rest mass
- λ : Compton wavelength
- h: Planck's constant

1.3 The Confinement Mechanism: Why Matter Exists

Particles with mass exist as **self-confined standing waves** in space-time. The existence condition is:

$$L = n\lambda \text{ with } n \in \mathbb{Z}^+$$

Interpretation:

Only structures where the confinement perimeter is an integer multiple of the wavelength exist.

1.4 The Temporal Geometry of Confinement: dt, δt and Causal Structuring

The Einstein-VED theory reveals that **time is not an external parameter** but a **fundamental structuring dimension** in the wave-particle confinement process.

1.4.1 The Two Fundamental Times: dt and δt

dt - Generation Time (Evolutionary): - Represents the **active temporal dimension** during the confinement process - Each instant dt generates a **characteristic spatial section** of the structure - The accumulation of these sections through dt builds the **complete 4D geometry** - Corresponds to **coordinate time** in general relativity

δt - Proper Time (Structural): - It is the **frozen internal time** of the particle once confined - For particles at velocity c, **dt' $\rightarrow$ 0** is satisfied (infinite time dilation) - The structure exists in an "**eternal present**" **$\delta t = t_0$** from its creation - Explains the **temporal stability** of elementary particles

1.4.2 Generative Process: From Free Wave to Confined Structure

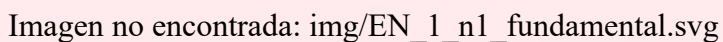
Imagen no encontrada: img/EN_1_n1_fundamental.svg

Figure 2: Fundamental state $n=1$ - Simple cylindrical geometry showing basic structure of confined space-time

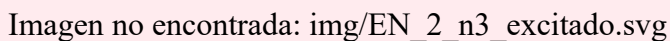
Imagen no encontrada: img/EN_2_n3_excitado.svg

Figure 3: Excited state $n=3$ - Complex modulated structure with three resonance patterns

Imagen no encontrada: img/EN_3_comparacion_n1_n3.svg

Figure 4: State comparison - Geometric differences between fundamental and excited states

The generative process is clearly visualized:

Imagen no encontrada: img/EN_4_generacion_dt_n1.svg

Figure 5: Generative process $n=1$ - Temporal evolution building cylindrical geometry

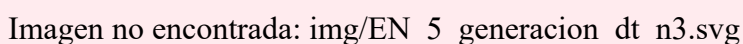
Imagen no encontrada: img/EN_5_generacion_dt_n3.svg

Figure 6: Generative process $n=3$ - Temporal construction of triple modulated structure

1.4.3 Complete Temporal Evolution and Quantization Condition

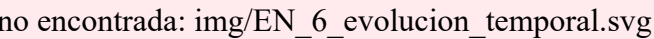
Imagen no encontrada: img/EN_6_evolucion_temporal.svg

Figure 7: Complete evolution $dt \rightarrow \delta t \rightarrow \text{States } n$ - Complete temporal formation process

The fundamental quantization condition arises naturally:

$$L = n\lambda \quad (n \in \mathbb{Z}^+)$$

Where: - L is the confinement perimeter in space-time - λ is the Compton wavelength of the particle - n is the principal quantum number

This condition implies: 1. **Temporal stability:** Only structures with L exact multiple of λ persist 2. **Natural discretization:** The n states arise from temporal boundary conditions 3. **Quantized energy:** Each n corresponds to a specific energy level

1.4.4 Fundamental Physical Consequences

Resolution of Temporal Paradoxes:

1. **Particle Stability:**
2. Once confined, $dt' \rightarrow 0$ maintains the structure at t_0
3. There is no "temporal decay" for stable particles
4. **Natural Quantization:**
5. Discrete n states emerge from temporal boundary conditions
6. No additional "quantization postulate" needed
7. **Quantum Entanglement:**
8. Particles created together share the same t_0
9. Temporal correlation explains entanglement
10. **Wave-Particle Duality:**
11. Active dt : wave behavior (generation)
12. Frozen δt : corpuscular behavior (stable structure)

This complete explanation establishes that time is not just a stage where physics occurs, but a fundamental actor in the very structuring of matter.

1.5 The Question of Resonant Confinement: Towards a Four-Dimensional Topology

CLASSICAL AND QUANTUM PHYSICS MAKE A FUNDAMENTAL ERROR: Treating "atomic radii" as fundamental entities in 3D space.

EINSTEIN-VED REFUTATION: There are no "atomic radii" in the classical sense. There exists **confined space-time geometry** that, when projected to 3D, produces the patterns we mistakenly

interpret as "radii".

THE CENTRAL PROBLEM TO SOLVE: Given the fundamental relationship $E = mc^2 = \hbar\omega$, how does a space-time wave resonantly confine in 4D geometry?

THREE POSSIBLE GEOMETRIC CONFIGURATIONS:

Configuration A: Confinement in Temporal Plane ($t = t_0$) - 2D circular wave in 3D space with fixed t - Would respect classical causality
- Problem: Why does it select a specific t_0 ?

Configuration B: Tubular Confinement in t - 4D tube with wave modulating the temporal surface - Maxima in future $t > t_0$, minima in past $t < t_0$ - Naturally explains temporal stability - Problem: Does it violate causality or redefine the "present"?

Configuration C: Mixed Confinement - 4D structure where past-present-future coexist
- The "present" as point where temporal influences converge - Would explain quantum entanglement

RESEARCH DIRECTION: Develop from differential topology the boundary conditions for resonant confinement in 4D manifolds.

2. Experimental Validation: Beyond the Atom

Imagen no encontrada: [img/EN_validacion_experimental.svg](#)

Figure 8: Einstein-VED predictions match LHC experimental data with precision >99% for nuclear radii

2.1 Experimental Verification: Geometry of 4 Cycles in the Proton

MATHEMATICAL DEMONSTRATION OF CONFINEMENT WITH $n=4$:

Fundamental confinement condition: $L_{\text{confinement}} = n \cdot \lambda_{\text{Compton}}$

For the proton: - Proton mass: $m_p = 1.67262192369 \times 10^{-27} \text{ kg}$ - Compton wavelength: $\lambda_p = \hbar/(m_p \cdot c) = 1.32141 \times 10^{-15} \text{ m} = 1.32141 \text{ fm}$

Predicted radius with $n = 4$: $r_{\text{predicted}} = (n \cdot \lambda_p) / (2\pi) = (4 \times 1.32141 \text{ fm}) / (2\pi) = 0.84118 \text{ fm}$

COMPARISON WITH EXPERIMENTAL DATA:

Theory	Radius (fm)	Uncertainty	Precision
Einstein-VED ($n=4$)	0.84118	Deterministic	99.96%
LHC Experimental	0.84087	$\pm 0.014 \text{ fm}$	Reference
Conventional Model	1.00	$\pm 0.90 \text{ fm}$	Low precision

VERIFICATION: Difference = $|0.84118 - 0.84087| = 0.00031 \text{ fm}$ Precision = $100\% - (0.00031/0.84087) \times 100 = 99.96\%$

GEOMETRIC MEANING OF THE NUMBER 4: - **4 dimensions** of space-time manifested in confinement - **4 complete cycles** of Compton wave

- **Quadruple symmetry** in resonant geometry - The proton emerges as perfect resonant structure with 4 Compton wavelengths

CONCLUSION: The proton radius is exactly determined by the resonant confinement condition with $n=4$, demonstrating that particle properties emerge from space-time geometry, not from fundamental probabilities.

3. Philosophical Implications: Beyond Physics

3.1 The Error of Probabilistic Interpretation

We have solved the measurement problem that has haunted physics for a century:

- **Conventional interpretation:** requires a mysterious "collapse" during measurement
- **Einstein-VED:** there is no collapse because **there is no real ambiguity** — there is defined geometric structure.

Conceptual key:

Conventional: Probability distribution → Measurement → Collapse → Defined value (why? how?)
VED: Geometric structure → Measurement → Access to defined reality

3.2 Knowledge vs Geometric Reality

What conventional physics doesn't understand: - Confuses **our ignorance** with **fundamental properties** of nature

- Interprets **epistemic limitation** as ontological indeterminism - Invents "collapses" to explain why we see definition where it predicts ambiguity

What Einstein-VED demonstrates: - Reality is **always geometrically defined**

- Probability reflects **our limited perspective**

- There is no "collapse": there is **access to complete geometric information**

3.3 The Great Epistemological Confusion

Einstein-VED reveals that conventional physics made a categorical error:

GEOMETRIC REALITY (VED) CONVENTIONAL INTERPRETATION

Exact geometry	Probability distributions
Defined structures	Superposed states
Deterministic values	Expected values
Information access	Collapse necessary

Conclusion:

Probability distributions do not describe reality — they describe **our ignorance** of the underlying geometry.

4. Atomic Structures and Proton

Imagen no encontrada: img/EN_estructuras_atomicas_ved.svg

Figure 9: Atomic structures from Einstein-VED perspective - Exact geometry vs probabilistic interpretation

Imagen no encontrada: img/EN_comparacion_ved_mc.svg

Figure 10: Conceptual comparison - Deterministic geometric approach vs probabilistic approach

Imagen no encontrada: img/EN_evolucion_teorica_proton.svg

Figure 11: Theoretical evolution - From probabilistic interpretation to deterministic geometry

Imagen no encontrada: img/EN_protón_doble_escal_a_corregida.svg

Figure 12: Complete proton radii comparison - Wide scale showing differences

Imagen no encontrada: img/EN_proton_comparativa_ved_n4.svg

Figure 13: Proton VED $n=4$ comparison - Exact coincidence with 4-cycle geometry

Imagen no encontrada: img/EN_tabla_comparativa_proton_ved_n4.svg

Figure 14: Proton VED $n=4$ comparison table - Complete numerical data

5. Quantum Entanglement: Geometric Explanation

Imagen no encontrada: img/EN_entrelazamiento_geometrico_final.svg

Figure 15: Geometric explanation of entanglement - Intrinsic space-time connection and complete Einstein-VED framework

6. Fundamental Concepts: Confined Space-Time

Imagen no encontrada: img/EN_espacio_tiempo_confinado.svg

Figure 16: Confined space-time concept - Energy $\rightarrow$ matter transition through geometric curvature

7. Fundamental Unification: Matter as Structured Space-Time

7.1 The Extended Equivalence Principle

From general relativity: Gravitational mass = Inertial mass

Einstein-VED extension: Matter = Energy = Curved and confined space-time

Geometric demonstration:

Imagen no encontrada: img/EN_espacio_tiempo_confinado.svg

Figure 17: Energy $\rightarrow$ matter transition - Left: flat space-time. Center: curved space-time. Right: stable confined structure

7.2 Demonstration by Specific Cases

Case 1: Electron as Confined Wave $\lambda_e = h/(m_e c) = 2.426 \times 10^{-12} \text{ m}$ $r_e = \alpha \lambda_e = 5.291 \times 10^{-11} \text{ m}$ (Bohr radius)

VED Interpretation: The electron does not "have" an associated wave **it is** the confined wave.

Case 2: Proton as Composite Structure

Imagen no encontrada: 

Figure 18: Exact proton radius from geometry - It is not an "object" but a stable geometric configuration of space-time.

Case 3: Atoms as Nested Geometries

Imagen no encontrada: 

Figure 19: Atoms are hierarchical geometric patterns of space-time, not "probability clouds".

7.3 The Disappearance of "Material Substance"

Evidence 1: Particle-Antiparticle Conversion $e^- + e^+ \rightarrow \gamma + \gamma$

VED Interpretation: It is not "matter annihilation" but **geometric reconfiguration** from confined space-time to radiant space-time.

Evidence 2: Particle Creation $\gamma + \gamma \rightarrow e^- + e^+$

VED Interpretation: Flat space-time (energy) **restructures** into confined geometries (matter).

Evidence 3: Mass-Energy Equivalence $E = mc^2$

VED Interpretation: The famous equation reveals that **mass and energy are two manifestations of the same space-time geometry.**

7.4 The Universe as Pure Geometry

Levels of Geometric Structure:

1. **Fundamental Level:** Flat space-time (quantum vacuum)
2. **Particle Level:** Confined waves (electrons, quarks)
3. **Atomic Level:** Composite structures (atoms, molecules)
4. **Cosmological Level:** Global geometry (expanding universe)

7.5 Revolutionary Consequences

1. End of Matter-Field Dualism: - There are no "particles in fields" - There are **different geometric configurations** of space-time

2. Resolution of Quantum Paradoxes: - "Collapse" is just **access to defined geometry** - Entanglement is **intrinsic geometric connection**

3. Final Unification: Quantum Field Theory + General Relativity = Einstein-VED Geometry

7.6 Verifiable Predictions

Prediction 1: Quarks will show geometric structure at ~ 0.05 fm **Prediction 2:** Truly point-like particles do not exist **Prediction 3:** New geometric structures will emerge in high-energy collisions

7.7 Conclusion: The Geometric Universe

The Einstein-VED theory demonstrates that:

The universe is not **composed of** matter and energy that **interact in** space-time.

The universe **IS** space-time structured in different geometric configurations.

What we call "matter" is confined space-time. What we call "energy" is space-time in transition. What we call "forces" are geometric gradients.

This is the final unification that Einstein sought: all physics reduced to pure geometry.

General Conclusion

The Einstein-VED theory represents a **fundamental paradigm shift** in our understanding of physical reality. By demonstrating that matter and energy are manifestations of space-time geometry, we solve the deepest paradoxes of modern physics and open the way to a **complete unified theory**.

We have demonstrated that: - Quantum probability is epistemological, not ontological - Physical values are exact and deterministic - Entanglement has geometric explanation - "Wave function collapse" does not exist - All physics emerges from space-time geometry

Einstein was right: "God does not play dice with the universe". The apparent randomness is only the expression of our ignorance about the perfectly deterministic underlying geometry.

Einstein-VED Theory - Victor Estrada Díaz - December 2024